

Assignment 1: Detecting Fraud and Intrusions in Imbalanced, Evolving Data

Core task	Classification and Anomaly Detection
Mode	Individual
Marks	100
Topics used	Data preprocessing, proximity measures, decision trees, k-NN, Naïve Bayes, ensembles, model evaluation

Background

Fraud detection and network intrusion detection are two of the most important real-world applications of classification. They are also among the hardest, because they break the assumptions made in textbook examples. The positive class is extremely rare (often well below 1%). Attackers and fraudsters change their behaviour, so the data distribution drifts over time. New attack types appear that were never labelled, so a purely supervised model cannot see them. Finally, analysts must be able to trust and explain every alert. In this assignment you will build, evaluate, and critically compare supervised classifiers and unsupervised anomaly detectors under these conditions.

Learning Outcomes

- Design a leakage-free preprocessing and evaluation pipeline for imbalanced data.
- Select metrics appropriate for rare-event detection and justify them.
- Compare supervised classification with unsupervised and semi-supervised anomaly detection.
- Evaluate robustness under temporal drift and previously unseen attack types.
- Explain individual model decisions to a non-technical analyst.

Dataset: choose ONE track

- **Track A — Financial fraud:** Credit Card Fraud Detection dataset (ULB Machine Learning Group, available on Kaggle). About 284,807 transactions with 492 frauds ($\approx 0.17\%$). Features V1–V28 are PCA-transformed; Time and Amount are raw.
- **Track B — IoT network intrusion:** CICIoT2023 dataset (Canadian Institute for Cybersecurity). Use a stratified sample of 200,000–500,000 flows that keeps all attack families. You will work with both the binary task (benign vs. attack) and the multi-class task (attack family).

Note: State the exact source, version and sampling procedure you used so your results can be reproduced.

Tasks

Task 1.1 — Data understanding and preprocessing (15 marks)

- Profile the data: shape, data types, missing values, duplicates, class distribution (bar chart and imbalance ratio), and distributions of the ten most informative features.
- Identify and handle infinite values, constant features, and highly correlated feature pairs ($|r| > 0.95$). Justify each removal.
- Apply scaling where needed and explain which of your classifiers are sensitive to scale and why. Link this to the proximity measures covered in class.
- Create two evaluation protocols: (i) a stratified random 70/15/15 train/validation/test split, and (ii) a time-ordered split (Track A: train on the earliest 70% of transactions by Time; Track B: see Task 1.5).
- Write a short “leakage audit”: list every place where test information could leak into training (scaling, oversampling, feature selection, duplicates) and show how your pipeline prevents it.

Task 1.2 — Supervised classification under imbalance (25 marks)

- Train a majority-class baseline and at least five classifiers: Decision Tree, k-NN, Naïve Bayes, Logistic Regression, Random Forest, and one gradient-boosting model (XGBoost or LightGBM).
- Tune hyperparameters using stratified 5-fold cross-validation on the training set only, optimising PR-AUC rather than accuracy.
- For the two best models, compare five imbalance strategies: (a) no treatment, (b) class weights, (c) random undersampling, (d) SMOTE, and (e) threshold moving on the validation set.
- SMOTE and undersampling must be applied inside the cross-validation folds (for example with `imblearn.pipeline.Pipeline`). Explain what goes wrong if they are applied before splitting.
- Present one consolidated results table: model $\times$ strategy $\times$ metrics.

Task 1.3 — Evaluating rare-event detectors correctly (15 marks)

- Report the confusion matrix, precision, recall, F1, Matthews Correlation Coefficient (MCC), ROC-AUC and PR-AUC for all final models.
- Show numerically why a model with 99.8% accuracy can be useless on this data.
- Plot ROC and precision–recall curves for the top three models on the same axes. Explain why the ROC curve can look optimistic under heavy imbalance.
- **Cost-sensitive decision:** define a cost matrix (Track A: a missed fraud costs the transaction Amount and a false alarm costs a fixed investigation cost, e.g., 5 units; Track B: justify your own costs). Choose the decision threshold that minimises total cost on the validation set and report the cost on the test set.
- Test whether the difference between your two best models is statistically significant (McNemar’s test on the test set, or a Wilcoxon signed-rank test over CV folds).

Task 1.4 — Unsupervised and semi-supervised anomaly detection (20 marks)

- Apply at least three anomaly detectors: Isolation Forest, Local Outlier Factor, and One-Class SVM (on a subsample if needed). Bonus (up to +5): add ECOD, COPOD or an autoencoder using PyOD.
- Run each detector in two settings: (i) unsupervised — fit on training data that includes anomalies; (ii) semi-supervised fit on normal records only.
- Labels may be used only for evaluation. Report PR-AUC, ROC-AUC and Precision@k, where k equals the number of true anomalies in the test set.
- **Label-budget experiment:** retrain your best supervised model using only 1%, 5%, 10%, 25% and 100% of the training labels (stratified). Plot PR-AUC against label budget with the anomaly detectors as horizontal reference lines. At what label budget does supervision start to win?

Task 1.5 — Drift and unseen threats (10 marks)

- **Track A:** divide the test period into at least four consecutive time windows. Plot recall and precision of the best supervised model and the best anomaly detector per window. Discuss any degradation.
- **Track B (zero-day simulation):** remove one or two attack families completely from training. Test the supervised model and the anomaly detectors on those unseen families. Report the detection rate on seen vs. unseen attacks.
- Propose (no implementation required) a monitoring and retraining strategy for deployment.

Task 1.6 — Explainability (10 marks)

- Report global feature importance for your best model using both impurity-based importance and permutation importance, and explain any disagreement.
- Use SHAP to explain three true positives, two false positives and one false negative. For each, write a two-line analyst-friendly explanation.
- Discuss whether the anomaly detector's top-ranked records agree with what the classifier considers important.

Task 1.7 — Discussion and recommendation (5 marks)

- In one page, recommend a detection system for a bank (Track A) or an IoT network operator (Track B). Should it be supervised, unsupervised, or a hybrid? Support every claim with your own results.

Marking Rubric — Assignment 1

Component	Marks	Criteria for full marks
1.1 Data understanding & preprocessing	15	Complete profiling; justified cleaning; both split protocols; convincing leakage audit.
1.2 Supervised classification	25	Six or more models including baseline; tuning inside CV; five imbalance strategies compared correctly.

Component	Marks	Criteria for full marks
1.3 Evaluation	15	All metrics; accuracy paradox shown; PR/ROC curves interpreted; cost-optimal threshold; significance test.
1.4 Anomaly detection	20	Three or more detectors in two settings; correct metrics; label-budget curve with clear conclusion.
1.5 Drift / unseen threats	10	Correct experimental design; clear evidence; realistic monitoring plan.
1.6 Explainability	10	Global and local explanations; errors explained; readable analyst summaries.
1.7 Recommendation	5	Evidence-based, practical, acknowledges limitations.